PROJECT AND TEAM INFORMATION

Project Title

SmartShield:Rule-Based Credit Card Fraud Detection Using Data Structures

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Credit card fraud has become a major problem in today digital payment world. As more people use credit cards and online payment services the chances of unauthorized transactions are also increasing. A single fraudulent transaction can cause financial loss to a customer and can also reduce their trust in online payment systems. Therefore detecting suspicious transactions quickly and accurately is very important for banks, merchants, and customers.

SmartShield takes a simpler approach. Instead of using AI or ML, it uses a rule-based system built entirely around fundamental Data Structures and Algorithms (DSA). This makes the system portable, transparent, and easy to understand. Every fraud alert is based on a specific rule so its decisions can be easily explained. SmartShield looks at important transaction details such as the amount, time, merchant, and location. It compares these details with the users normal spending habits. If a transaction is very different from the users usual behaviour the system can mark it as suspicious and generate an alert. By comparing these factors with a users usual spending behavior it can identify transactions that appear unusual or suspicious.

The system also makes effective use of different data structures. Hash Table provide quick access to transaction histories. Dequeue help track recent transactions. Sorting help identify unusual patterns and outliers and Graphs represent connections between users merchants and locations.

Overall SmartShield shows that fraud detection does not always need AI or ML techniques. By using suitable Data Structures and Algorithms, it is possible to create a fast, simple, understandable, and practical system that can help protect users from fraudulent credit card transactions.

State of the Art / Current solution

Today, banks use automated fraud detection systems to keep customers and their transactions safe. These systems check different details of a transaction such as the amount, location, merchant, transaction frequency and the users previous spending habits. By comparing a new transaction with normal spending pattern the system can identify unusual or suspicious activity.

SmartShield uses a simple approach that is mainly designed for academic learning and display.

Unlike modern fraud detection systems that depend on AI or Machine Learning SmartShield uses basic Data Structures and Algorithms (DSA) to manage transaction data and identify suspicious activities.This makes the system easy to understand use and explain. Every step of the fraud detection process can be followed clearly which helps students understand how the system works. It also shows how basic computer science concepts can be used to solve a real-world problem such as credit card fraud.

Project Goals and Milestones

The main goal of this project is to build a simple terminal-based credit card fraud detection system that shows how Data Structures and Algorithms (DSA) can be used to solve a real-world problem. This is not meant to be a commercial or production-ready product it is a prototype built mainly to show how core DSA concepts work in practice. To keep things organized and easy to manage the project is split into three phases.

Phase 1: Planning & Requirement Analysis

This initial phase lays the groundwork for the entire project. It involves clearly defining the problem statement setting out the projects objectives and scope and identifying the specific requirements the system needs to fulfill. This phase also includes deciding on the technology stack specifically selecting C++ as the programming language and determining which data structures will best support the system functionality.

Phase 2: System Design & Development

With the requirements in place this phase shifts focus to actually building the system. It involves designing the overall system architecture mapping out the logical flow through flowcharts and planning how transaction data will be handled and stored. The core fraud detection logic is then implemented using C++ the chosen data structures to analyze transactions and flag suspicious behavior.

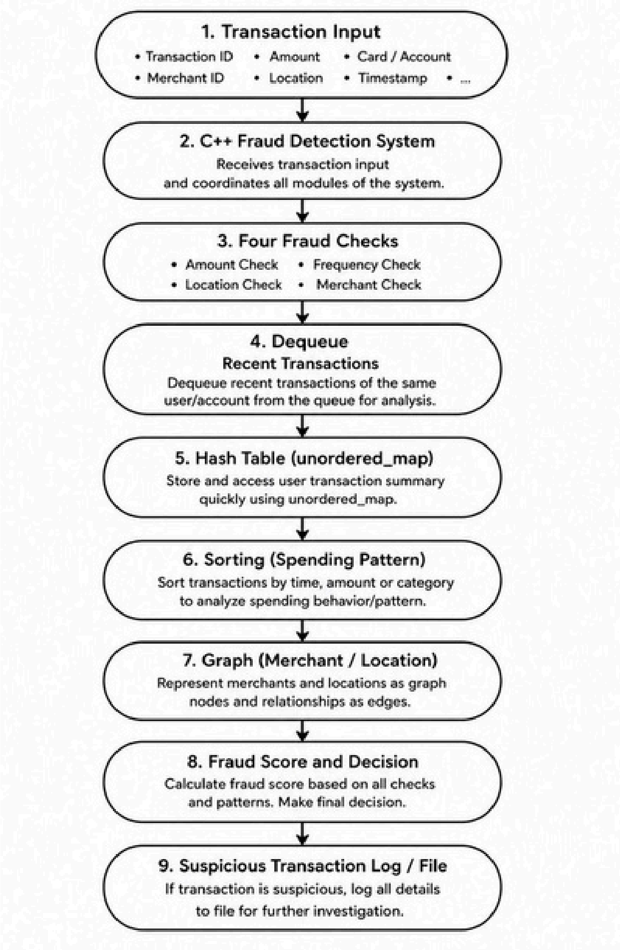
Phase 3: Testing & Final Implementation

The final phase centers on validating the system performance. This includes running the system against a variety of transaction scenarios to confirm it can accurately identify fraudulent cases. Any bugs or issues uncovered during testing are addressed and resolved. Once the system is stable sample reports are generated to showcase its output and final documentation is prepared alongside a demonstration to present the completed project.

Project Approach

The Credit Card Fraud Detection System will be developed using C++ and Data Structures, with theaim of creating a simple system that can efficiently store process and analyze transaction records. C++ will be used to develop the complete application while different Data Structures will be selected based on the type of operation being performed. Structures and Classes will store important transaction details such as Transaction ID, amount, date, time, location and transaction status. Arrays or vectors will be used to manage multiple transaction records while Queue will process incoming transactions in the order they are received. A Linked List can be used when transaction records need to be added or removed dynamically and a Stack can maintain the most recent transaction history. To make transaction management faster searching algorithms will be used to locate specific transactions while sorting algorithms will organize transactions based on factor such as amount time or risk score. Hashing may also be used to quickly find transactions using their Transaction IDs. Pointers will be used where required particularly for dynamic structures such as linked lists.The system will be divided into separate C++ functions or modules for transaction input processing fraud detection searching sorting and report generation. File handling will allow transaction records and fraud-detection results to be saved and retrieved when needed.Finally the fraud detection module will examine transaction details and identify suspicious activity using predefined conditions or calculated risk score. Based on the analysis each transaction will be classified as Legitimate or Fraudulent and the result will be displayed to the user.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The SmartShield Credit Card Fraud Detection System offers a simple practical way to spot suspicious transactions using C++ paired with fundamental Data Structures and Algorithms. It stores organizes searches and sorts transaction records while examining key details like amount, time, merchant, and location. Based on predefined rules and calculated risk scores each transaction is classified as either Legitimate or Fraudulent with file handling used to save both the data and results. Ultimately SmartShield shows how core DSA concepts can be applied to a real world challenge resulting in a lightweight transparent and easy-to-understand fraud detection system.

Assumptions

The SmartShield project works on a few basic assumptions to keep fraud detection simple and easy to manage.Every transaction is assumed to have correct and complete detail like the card ID, transaction amount, time, merchant, and location. The system looks at a users former transaction to understand their normal spending habits. It assumes the location data is fairly accurate. This help SmartShield spot transactions happening from unusual place compared to where the user normally shops. Each transaction gets a fraud score based on a set of rules. If this score reaches 2 or more higher the transaction is marked as suspicious and flagged for further review. Since SmartShield is mainly a academic project its main goal is to show how Data Structures and Algorithms can be applied to fraud detection. It is not meant to replace the advanced real-world fraud detection systems used by banks and financial institutions.

References

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